

# Supplementary materials

Predicting emotional valence in autism: a preregistered study in the Bayesian Brain framework

Plank et al.

2026-04-25

## S1 Package versions

The following packages are used in this RMarkdown file:

```
## [1] "R version 4.5.2 (2025-10-31)"
## [1] "knitr version 1.51"
## [1] "ggplot2 version 4.0.2"
## [1] "brms version 2.23.0"
## [1] "designr version 0.1.13"
## [1] "bridgesampling version 1.2.1"
## [1] "tidyverse version 2.0.0"
## [1] "ggpubr version 0.6.2"
## [1] "ggpattern version 1.2.1"
## [1] "correlation version 0.8.8"
## [1] "gggrain version 0.1.0"
## [1] "bayesplot version 1.15.0"
## [1] "SBC version 0.5.0.9000"
## [1] "rstatix version 0.7.3"
## [1] "BayesFactor version 0.9.12.4.7"
## [1] "effectsize version 1.0.1"
## [1] "bayestestR version 0.17.0"
```

## S2 Methods

### S2.1 Preprocessing of pupil sizes

We classified any extreme values (above 8 and below 2mm) as well as participant-specific extreme velocities and blinks as artefacts. Then, we low-pass filtered the data with a threshold of 4Hz based on Kret and Sjak-Shie (2019), smoothed the data and interpolated all artefacts shorter than half a second with cubic spline interpolation. Next, data was downsampled to 100Hz and trials with more than one third of the datapoints missing were excluded from analysis. Last, we applied correction based on a baseline of 200ms before onset of the face presentation. We visualised data quality and excluded trials if the pupil trace indicates recording artefacts or baseline artefacts as well as if the baseline pupil size is considered an outlier (absolute z-score larger than 2).

### S2.2 Bayesian linear (mixed) models

All contrasts are set to sum coding. The models for the volatilities extracted from the HGF and for the pupil sizes used a Gaussian likelihood, while both the reaction times and learning rate updates were modelled using a lognormal distribution. The reaction time model additionally included a shift. All models included the predictor Group (ASD, COMP). The reaction time model also included Expectancy, Phase and Difficulty as well as all interactions on the population-level, and participant with slopes for Expectancy, Phase, Difficulty and their interactions as well as trials with slopes for Group on the group-level. The hypothesis regarding pupil sizes was evaluated using a model with the predictors Group, Expectancy and their interaction as well as intercepts for the participants on the group-level. Since pupil sizes are sensitive to physical movements (Mathôt & Vilotijević, 2023), we include RT of each trial instead of including trial on the group-level, as preregistered. The model for the learning rate updates included Group, Level (environmental or cue-outcome association) and Change (stable to volatile or volatile to stable) on the population-level as well as participants with slopes for Level and Change on the group-level.

## S3 Model evaluation

### S3.1 Parameter recovery and model identifiability

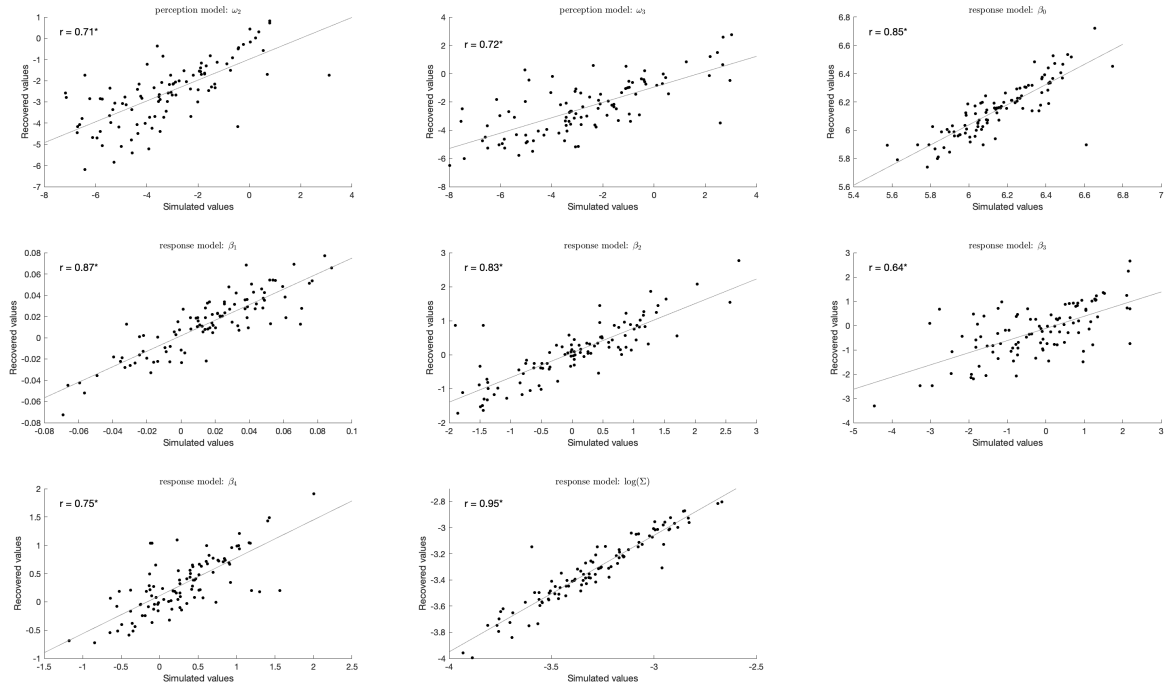


Figure S1: Enhanced HGF with empirical priors

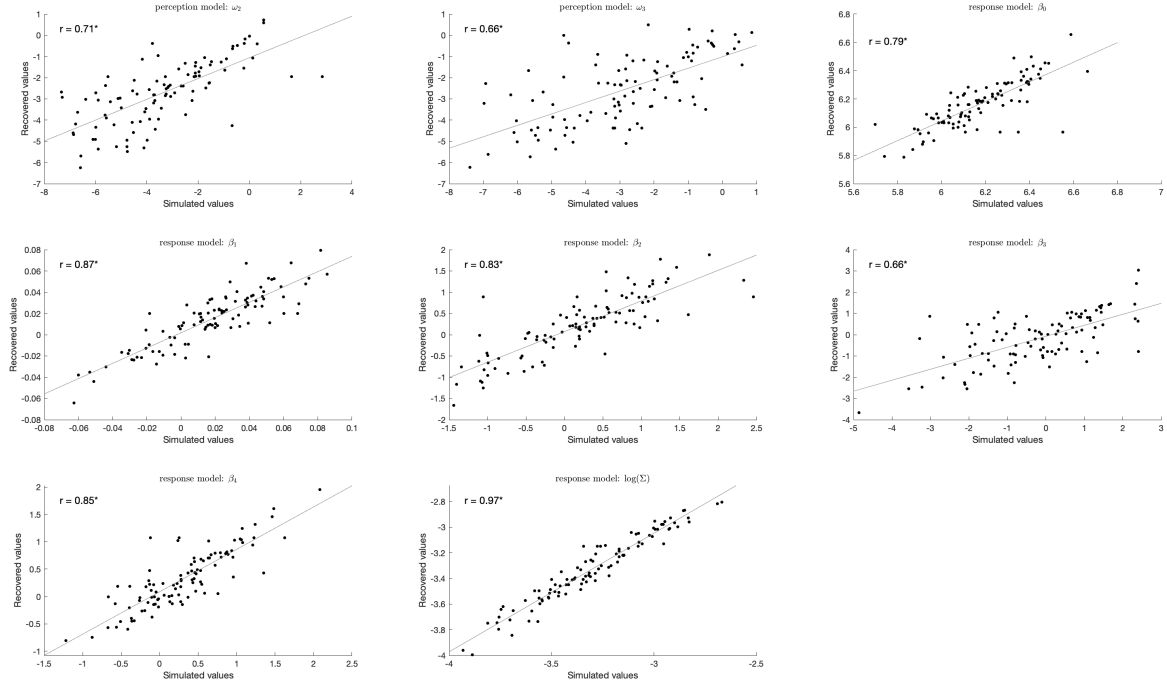


Figure S2: Original HGF with empirical priors

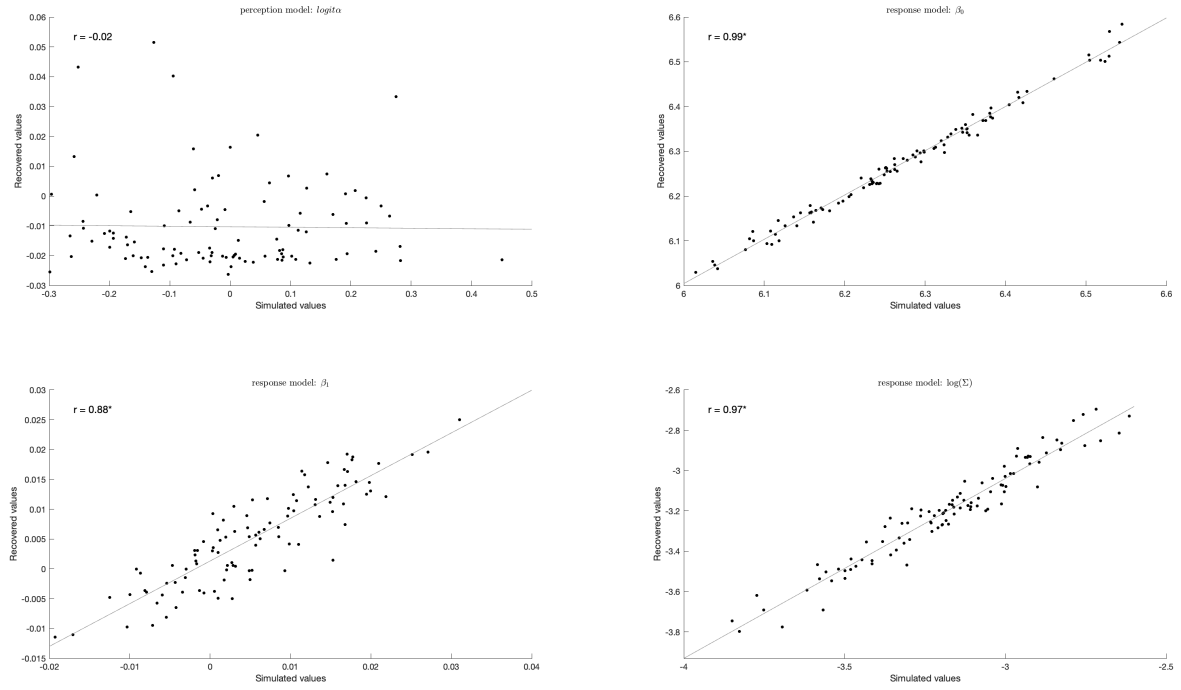


Figure S3: Rescorla-Wagner with empirical priors

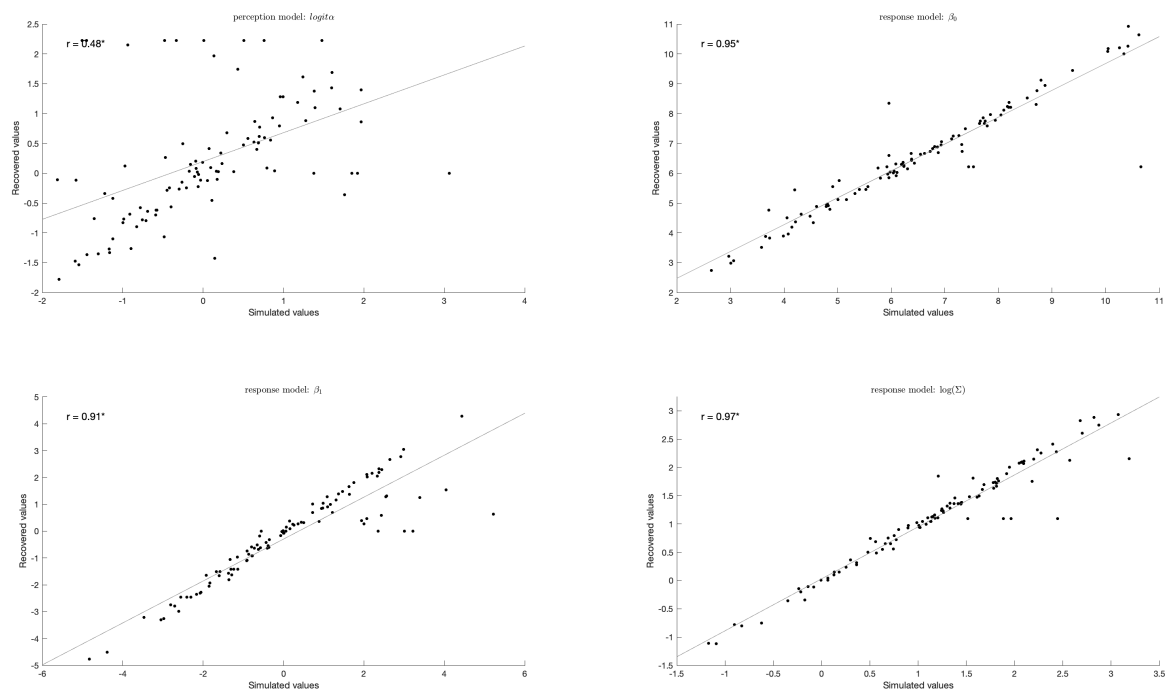


Figure S4: Rescorla-Wagner with default priors

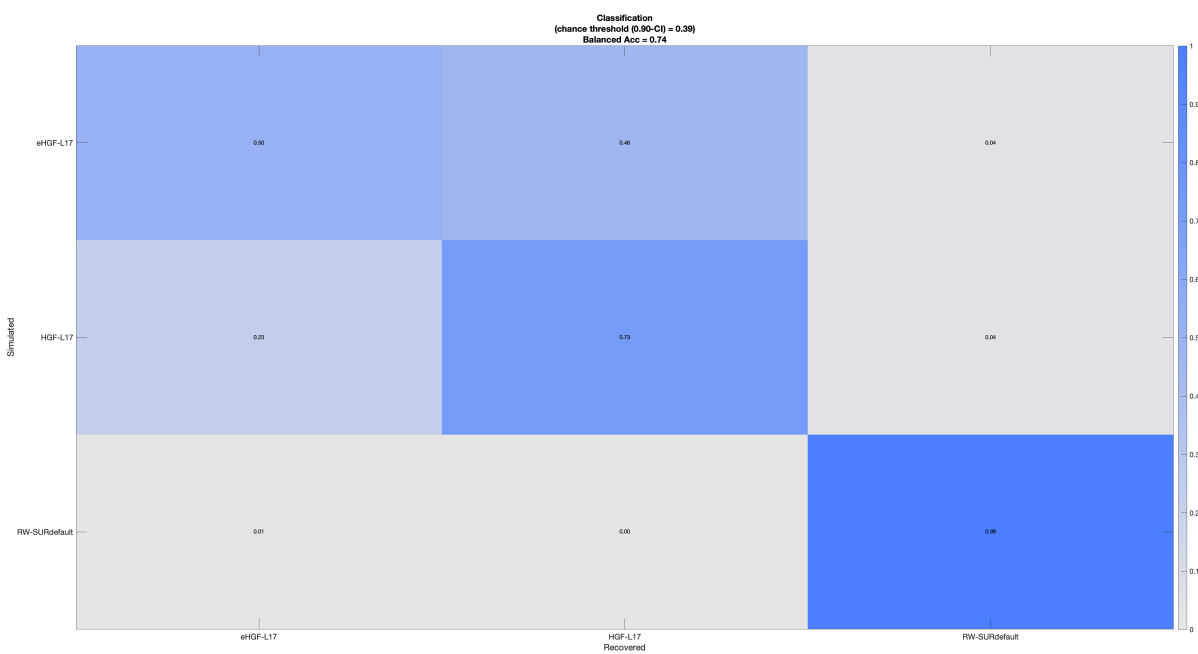


Figure S5: Model identifiability

### S3.2 Posterior predictive checks for oHGF

Here, we compare simulated and observed reaction times. The simulated reaction times were simulated based on the model which best fit the data, namely the original HGF (oHGF).

Table S1: Within-condition correlations between median values of simulated and observed reaction times

| Condition                          | rho   | CI_low | CI_high | pd | BF           | n_Obs |
|------------------------------------|-------|--------|---------|----|--------------|-------|
| volatile - expected - easy         | 0.936 | 0.898  | 0.965   | 1  | 2.083329e+18 | 44    |
| volatile - expected - medium       | 0.904 | 0.833  | 0.947   | 1  | 9.580647e+14 | 44    |
| prevolatile - expected - hard      | 0.896 | 0.835  | 0.951   | 1  | 1.587879e+14 | 44    |
| postvolatile - expected - medium   | 0.890 | 0.812  | 0.939   | 1  | 9.217609e+13 | 44    |
| postvolatile - expected - easy     | 0.890 | 0.817  | 0.942   | 1  | 8.381426e+13 | 44    |
| prevolatile - expected - medium    | 0.876 | 0.791  | 0.927   | 1  | 7.588768e+12 | 44    |
| prevolatile - expected - easy      | 0.863 | 0.749  | 0.935   | 1  | 1.457185e+12 | 44    |
| volatile - expected - hard         | 0.862 | 0.778  | 0.930   | 1  | 1.587908e+12 | 44    |
| postvolatile - expected - hard     | 0.855 | 0.754  | 0.920   | 1  | 5.547958e+11 | 44    |
| volatile - unexpected - easy       | 0.757 | 0.619  | 0.859   | 1  | 5.393531e+07 | 44    |
| postvolatile - unexpected - easy   | 0.745 | 0.601  | 0.864   | 1  | 2.665158e+07 | 44    |
| prevolatile - unexpected - medium  | 0.744 | 0.620  | 0.859   | 1  | 2.236104e+07 | 44    |
| prevolatile - unexpected - hard    | 0.738 | 0.591  | 0.849   | 1  | 1.654123e+07 | 44    |
| prevolatile - unexpected - easy    | 0.683 | 0.503  | 0.815   | 1  | 5.793616e+05 | 44    |
| volatile - unexpected - hard       | 0.679 | 0.501  | 0.810   | 1  | 3.980486e+05 | 44    |
| postvolatile - unexpected - medium | 0.670 | 0.486  | 0.804   | 1  | 2.616542e+05 | 44    |
| postvolatile - unexpected - hard   | 0.617 | 0.434  | 0.777   | 1  | 2.473718e+04 | 44    |
| volatile - unexpected - medium     | 0.583 | 0.367  | 0.749   | 1  | 5.075359e+03 | 44    |

Table S2: Within-condition mean and SDs of observed and simulated RTs

| phase        | expected   | difficulty | mean observed | mean simulated | SD observed | SD simulated |
|--------------|------------|------------|---------------|----------------|-------------|--------------|
| prevolatile  | expected   | easy       | 544.2         | 594.4          | 78.6        | 93.1         |
| prevolatile  | expected   | medium     | 575.5         | 593.4          | 79.0        | 84.7         |
| prevolatile  | expected   | hard       | 587.7         | 598.5          | 88.7        | 94.5         |
| prevolatile  | unexpected | easy       | 648.8         | 649.5          | 138.3       | 120.2        |
| prevolatile  | unexpected | medium     | 650.9         | 640.4          | 116.8       | 119.4        |
| prevolatile  | unexpected | hard       | 644.6         | 631.3          | 125.2       | 105.7        |
| volatile     | expected   | easy       | 546.2         | 598.5          | 74.1        | 88.3         |
| volatile     | expected   | medium     | 553.0         | 603.2          | 73.8        | 92.4         |
| volatile     | expected   | hard       | 590.7         | 608.9          | 83.9        | 87.6         |
| volatile     | unexpected | easy       | 598.4         | 610.4          | 84.4        | 90.2         |
| volatile     | unexpected | medium     | 584.8         | 596.8          | 91.6        | 80.0         |
| volatile     | unexpected | hard       | 609.5         | 600.7          | 107.0       | 102.7        |
| postvolatile | expected   | easy       | 552.9         | 593.6          | 80.1        | 87.7         |
| postvolatile | expected   | medium     | 566.2         | 587.2          | 74.5        | 83.2         |
| postvolatile | expected   | hard       | 605.1         | 591.3          | 100.0       | 92.0         |
| postvolatile | unexpected | easy       | 570.6         | 607.0          | 123.9       | 92.6         |
| postvolatile | unexpected | medium     | 565.0         | 617.6          | 113.0       | 96.0         |
| postvolatile | unexpected | hard       | 595.5         | 614.7          | 100.7       | 97.2         |

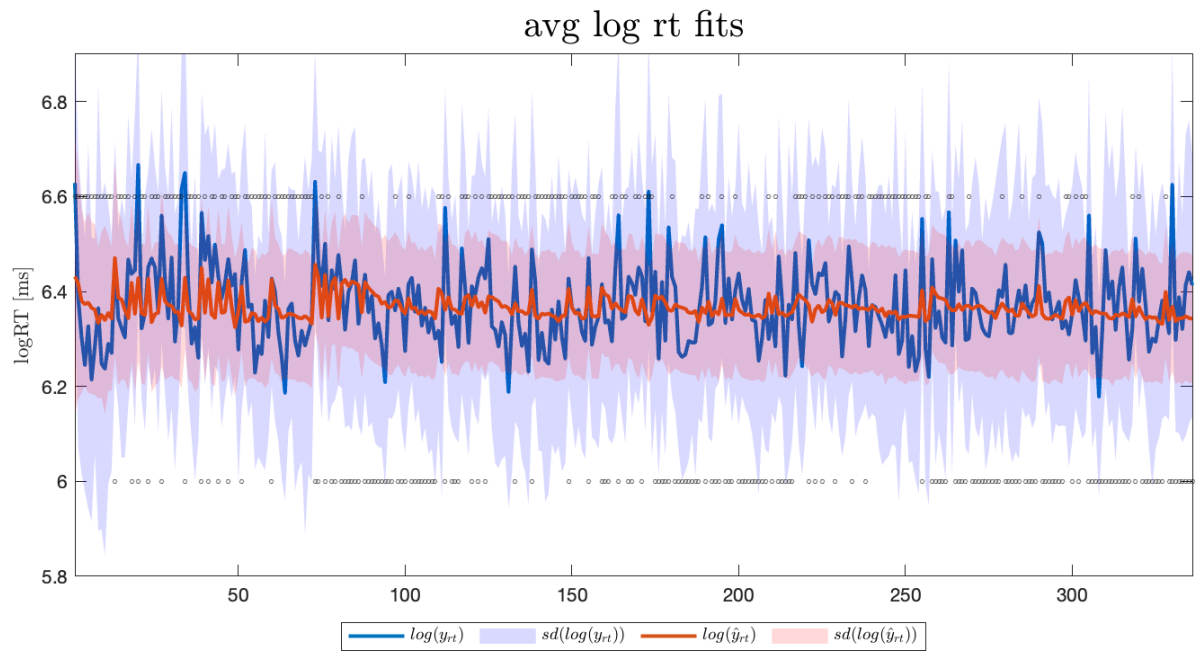


Figure S6: Task development of average  $\log(\text{rt})$  and simulated  $\log(\text{rt})$  as well as response model parameters

Table S3: Posterior predictive checks: comparison of model based on simulated and model based on observed data

| effect                                    | overlap | estimate based on observed RT | estimate based on simulated RT |
|-------------------------------------------|---------|-------------------------------|--------------------------------|
| b_diagnosis1                              | 0.9279  | 0.0304                        | 0.0308                         |
| b_diagnosis1:expected1                    | 0.8955  | -0.0013                       | -0.0022                        |
| b_diagnosis1:expected1:phase2             | 0.8023  | 0.0002                        | 0.0004                         |
| b_diagnosis1:phase2                       | 0.7896  | -0.0007                       | -0.0028                        |
| b_diagnosis1:difficulty1                  | 0.7698  | 0.0008                        | 0.0014                         |
| b_diagnosis1:phase1                       | 0.7657  | 0.0000                        | 0.0061                         |
| b_diagnosis1:expected1:difficulty1        | 0.7484  | -0.0013                       | 0.0004                         |
| b_phase2                                  | 0.7445  | -0.0071                       | -0.0077                        |
| b_phase1                                  | 0.6953  | 0.0269                        | 0.0177                         |
| b_diagnosis1:phase2:difficulty2           | 0.6822  | 0.0059                        | 0.0028                         |
| b_diagnosis1:phase2:difficulty1           | 0.6646  | -0.0025                       | 0.0009                         |
| b_expected1:phase1:difficulty1            | 0.6469  | -0.0078                       | -0.0041                        |
| b_diagnosis1:expected1:phase1:difficulty2 | 0.6183  | -0.0005                       | -0.0053                        |
| b_diagnosis1:expected1:phase1:difficulty1 | 0.6142  | 0.0034                        | 0.0082                         |
| b_diagnosis1:expected1:phase2:difficulty1 | 0.6136  | 0.0010                        | -0.0033                        |
| b_expected1:difficulty2                   | 0.6099  | 0.0020                        | -0.0011                        |
| b_expected1:phase1:difficulty2            | 0.6069  | 0.0075                        | 0.0022                         |
| b_phase1:difficulty1                      | 0.5789  | -0.0022                       | 0.0049                         |
| b_phase2:difficulty2                      | 0.5421  | -0.0110                       | -0.0041                        |
| b_phase2:difficulty1                      | 0.5313  | 0.0077                        | 0.0002                         |
| b_difficulty2                             | 0.5261  | -0.0086                       | -0.0025                        |
| b_expected1:phase2:difficulty1            | 0.5175  | -0.0002                       | -0.0081                        |
| b_expected1:phase2:difficulty2            | 0.5034  | -0.0037                       | 0.0048                         |
| b_diagnosis1:difficulty2                  | 0.4636  | -0.0040                       | 0.0011                         |
| b_diagnosis1:expected1:phase2:difficulty2 | 0.4329  | -0.0047                       | 0.0024                         |
| b_diagnosis1:expected1:difficulty2        | 0.4211  | 0.0025                        | -0.0034                        |
| b_diagnosis1:phase1:difficulty2           | 0.4066  | -0.0067                       | 0.0019                         |
| b_diagnosis1:expected1:phase1             | 0.4048  | 0.0052                        | -0.0024                        |
| b_expected1:difficulty1                   | 0.3869  | -0.0146                       | -0.0050                        |
| b_diagnosis1:phase1:difficulty1           | 0.3710  | 0.0062                        | -0.0033                        |
| b_phase1:difficulty2                      | 0.3059  | 0.0174                        | 0.0008                         |
| b_expected1:phase1                        | 0.2467  | -0.0374                       | -0.0222                        |
| b_expected1                               | 0.1858  | -0.0361                       | -0.0228                        |
| b_expected1:phase2                        | 0.0314  | 0.0013                        | 0.0245                         |
| b_difficulty1                             | 0.0076  | -0.0284                       | 0.0036                         |

### S3.3 Comparison of LME across groups

```
## ---
## Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis : fixed
```

|           | bf       | error    | time                     | code          |
|-----------|----------|----------|--------------------------|---------------|
| diagnosis | -1.04496 | 5.81e-05 | Wed Oct 15 12:08:23 2025 | 144d96dd874c4 |

## S4 Volatility parameters and learning rate updates

### S4.1 Phasic volatility

Phasic volatility was modeled with a Gaussian linear model with one predictor group (ASD = 1, COMP = -1).

Table S5: Estimates for the Phasic volatility model

|            | Estimate | Est.Error | Q2.5   | Q97.5 |
|------------|----------|-----------|--------|-------|
| Intercept  | 0.154    | 0.072     | 0.009  | 0.299 |
| diagnosis1 | 0.101    | 0.070     | -0.034 | 0.238 |

## S4.2 Environmental tonic volatility

Modelled with a Gaussian linear model with one predictor group (ASD = 1, COMP = -1).

Table S6: Estimates for the Environmental tonic volatility model

|            | Estimate | Est.Error | Q2.5   | Q97.5  |
|------------|----------|-----------|--------|--------|
| Intercept  | -2.247   | 0.222     | -2.682 | -1.811 |
| diagnosis1 | -0.034   | 0.165     | -0.357 | 0.292  |

## S4.3 Bernoulli model with HGF parameters

We also used an explorative Bernoulli model containing all scaled HGF parameters to predict group membership (ASD, COMP).

Table S7: Estimates for the explorative model including all HGF parameters

|           | Estimate | Est.Error | Q2.5   | Q97.5 |
|-----------|----------|-----------|--------|-------|
| Intercept | 0.000    | 0.272     | -0.521 | 0.531 |
| sbe1      | -0.460   | 0.420     | -1.297 | 0.355 |
| sbe2      | 0.282    | 0.399     | -0.483 | 1.085 |
| sbe3      | 0.278    | 0.457     | -0.593 | 1.195 |
| sbe4      | 0.806    | 0.432     | -0.021 | 1.679 |
| sze       | 0.240    | 0.402     | -0.551 | 1.035 |
| som2      | 0.302    | 0.351     | -0.380 | 1.000 |
| som3      | -0.139   | 0.352     | -0.842 | 0.542 |

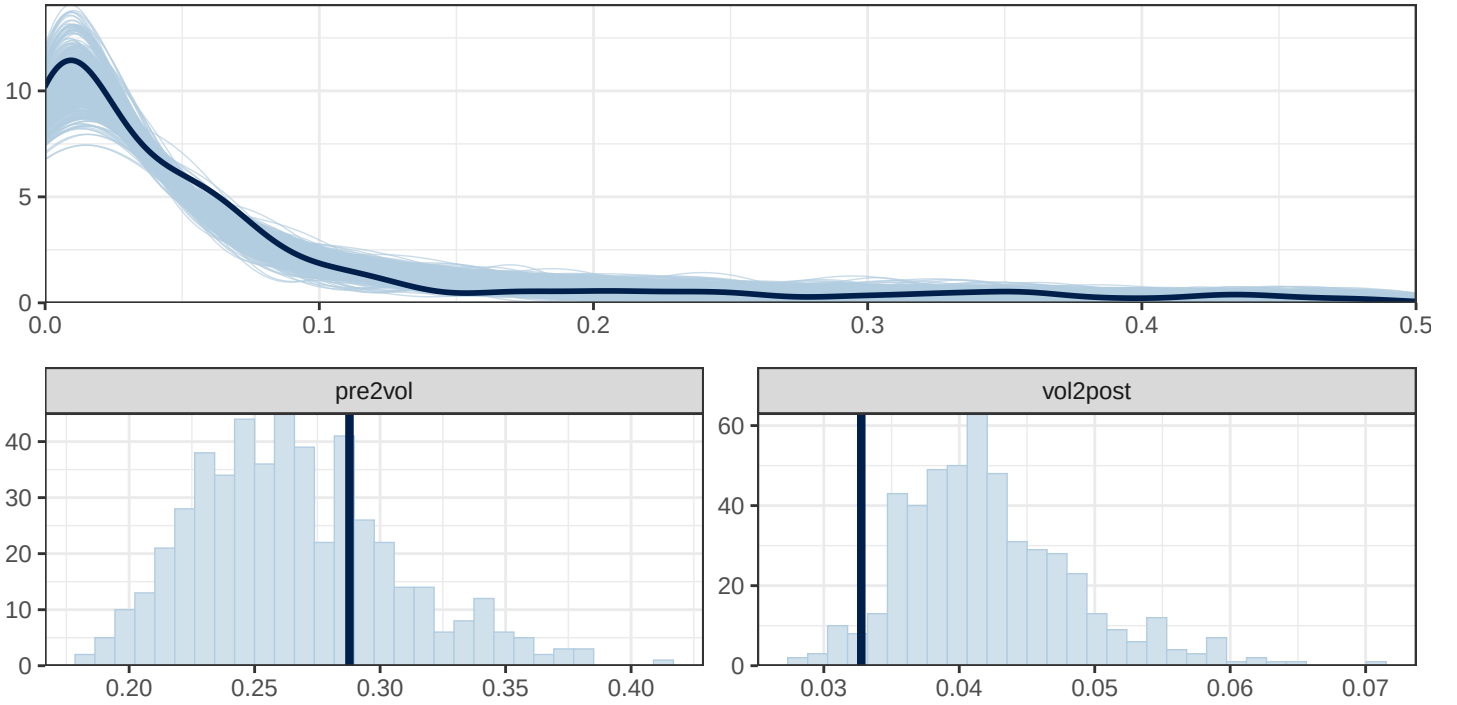
## S4.4 Learning rate updates

Modelled with a lognormal linear mixed model with the predictors group (ASD, COMP), level (second- or third-level alpha) and change (prevolatile to volatile or volatile to postvolatile phase) and the random intercept for subject and slopes for level and change.

Table S8: Estimates for the Learning rate updates model

|                           | Estimate | Est.Error | Q2.5   | Q97.5  |
|---------------------------|----------|-----------|--------|--------|
| Intercept                 | -3.575   | 0.083     | -3.742 | -3.409 |
| diagnosis1                | 0.168    | 0.085     | 0.001  | 0.333  |
| level1                    | 1.358    | 0.119     | 1.126  | 1.596  |
| change1                   | 1.109    | 0.045     | 1.017  | 1.198  |
| diagnosis1:level1         | 0.029    | 0.121     | -0.207 | 0.273  |
| diagnosis1:change1        | -0.090   | 0.045     | -0.178 | -0.001 |
| level1:change1            | -0.240   | 0.041     | -0.320 | -0.159 |
| diagnosis1:level1:change1 | 0.118    | 0.041     | 0.037  | 0.198  |



**Figure S7: Posterior predictive checks****Table S9: ANOVA of ranked learning rate updates**

|                                                                                                         | bf      |
|---------------------------------------------------------------------------------------------------------|---------|
| level + change                                                                                          | 105.308 |
| level + change + level:change                                                                           | 105.274 |
| diagnosis + level + change + level:change                                                               | 104.709 |
| diagnosis + level + change                                                                              | 104.631 |
| diagnosis + level + change + diagnosis:change                                                           | 103.743 |
| diagnosis + level + change + diagnosis:change + level:change                                            | 103.633 |
| diagnosis + level + diagnosis:level + change + level:change                                             | 103.524 |
| diagnosis + level + diagnosis:level + change                                                            | 103.471 |
| diagnosis + level + diagnosis:level + change + diagnosis:change + level:change                          | 102.352 |
| diagnosis + level + diagnosis:level + change + diagnosis:change                                         | 102.351 |
| diagnosis + level + diagnosis:level + change + diagnosis:change + level:change + diagnosis:level:change | 101.547 |
| level                                                                                                   | 51.219  |
| diagnosis + level                                                                                       | 49.990  |
| diagnosis + level + diagnosis:level                                                                     | 48.643  |
| change                                                                                                  | 23.304  |
| diagnosis + change                                                                                      | 21.886  |
| diagnosis + change + diagnosis:change                                                                   | 20.537  |
| diagnosis                                                                                               | -1.494  |

## S5 Reaction times, pupil sizes and accuracies

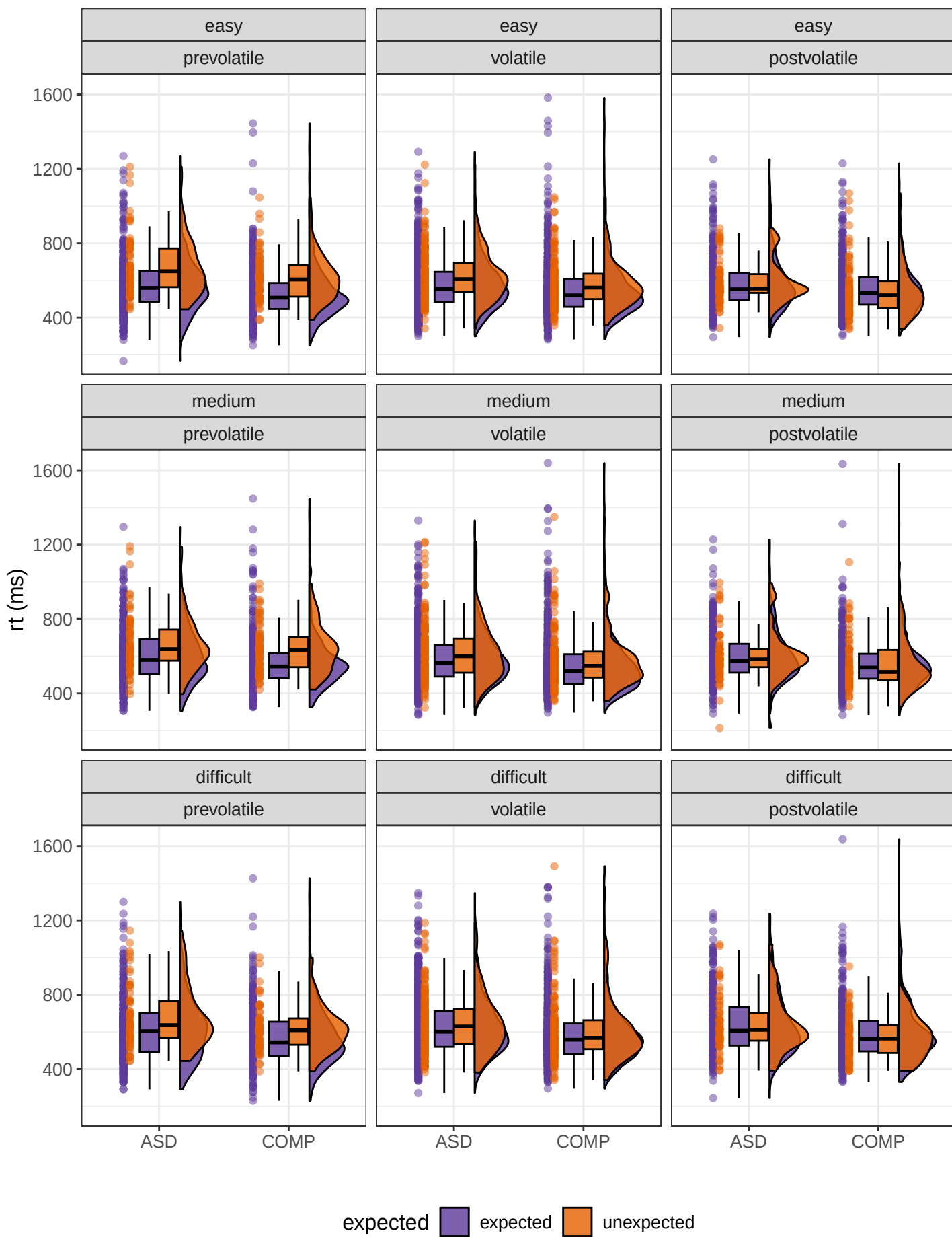
### S5.1 Reaction times

Reaction times were modelled with a shifted lognormal distribution and the predictors group (ASD, COMP), expected (yes, no), phase (prevolatile, volatile, postvolatile), difficulty (easy, medium, hard) and their interactions. All predictors were sum-coded. Furthermore, we added random intercept for subjects (slopes: expected, phase, difficulty and their interactions) and trials (slope: diagnosis). The full model can be downloaded from OSF: <https://osf.io/wkgec>.

Table S10: Fixed effects of the reaction time model

|                                         | Estimate | Est.Error | Q2.5   | Q97.5  |
|-----------------------------------------|----------|-----------|--------|--------|
| Intercept                               | 6.174    | 0.029     | 6.117  | 6.230  |
| diagnosis1                              | 0.030    | 0.021     | -0.010 | 0.070  |
| expected1                               | -0.036   | 0.006     | -0.048 | -0.024 |
| phase1                                  | 0.027    | 0.012     | 0.004  | 0.050  |
| phase2                                  | -0.007   | 0.008     | -0.022 | 0.008  |
| difficulty1                             | -0.028   | 0.008     | -0.044 | -0.013 |
| difficulty2                             | -0.009   | 0.008     | -0.024 | 0.007  |
| diagnosis1:expected1                    | -0.001   | 0.004     | -0.009 | 0.006  |
| diagnosis1:phase1                       | 0.000    | 0.010     | -0.019 | 0.019  |
| diagnosis1:phase2                       | -0.001   | 0.005     | -0.011 | 0.010  |
| expected1:phase1                        | -0.037   | 0.009     | -0.054 | -0.020 |
| expected1:phase2                        | 0.001    | 0.007     | -0.013 | 0.015  |
| diagnosis1:difficulty1                  | 0.001    | 0.005     | -0.008 | 0.010  |
| diagnosis1:difficulty2                  | -0.004   | 0.005     | -0.013 | 0.005  |
| expected1:difficulty1                   | -0.015   | 0.008     | -0.030 | 0.001  |
| expected1:difficulty2                   | 0.002    | 0.008     | -0.014 | 0.018  |
| phase1:difficulty1                      | -0.002   | 0.011     | -0.024 | 0.020  |
| phase2:difficulty1                      | 0.008    | 0.010     | -0.012 | 0.027  |
| phase1:difficulty2                      | 0.017    | 0.011     | -0.005 | 0.040  |
| phase2:difficulty2                      | -0.011   | 0.010     | -0.030 | 0.008  |
| diagnosis1:expected1:phase1             | 0.005    | 0.005     | -0.005 | 0.015  |
| diagnosis1:expected1:phase2             | 0.000    | 0.004     | -0.008 | 0.008  |
| diagnosis1:expected1:difficulty1        | -0.001   | 0.005     | -0.010 | 0.008  |
| diagnosis1:expected1:difficulty2        | 0.003    | 0.005     | -0.007 | 0.012  |
| diagnosis1:phase1:difficulty1           | 0.006    | 0.007     | -0.007 | 0.019  |
| diagnosis1:phase2:difficulty1           | -0.002   | 0.006     | -0.014 | 0.010  |
| diagnosis1:phase1:difficulty2           | -0.007   | 0.007     | -0.020 | 0.006  |
| diagnosis1:phase2:difficulty2           | 0.006    | 0.006     | -0.005 | 0.017  |
| expected1:phase1:difficulty1            | -0.008   | 0.011     | -0.029 | 0.014  |
| expected1:phase2:difficulty1            | 0.000    | 0.010     | -0.019 | 0.019  |
| expected1:phase1:difficulty2            | 0.008    | 0.011     | -0.015 | 0.030  |
| expected1:phase2:difficulty2            | -0.004   | 0.010     | -0.023 | 0.016  |
| diagnosis1:expected1:phase1:difficulty1 | 0.003    | 0.007     | -0.010 | 0.017  |
| diagnosis1:expected1:phase2:difficulty1 | 0.001    | 0.006     | -0.010 | 0.012  |
| diagnosis1:expected1:phase1:difficulty2 | 0.000    | 0.007     | -0.014 | 0.013  |
| diagnosis1:expected1:phase2:difficulty2 | -0.005   | 0.006     | -0.016 | 0.006  |

Figure S8: Reaction times per subject



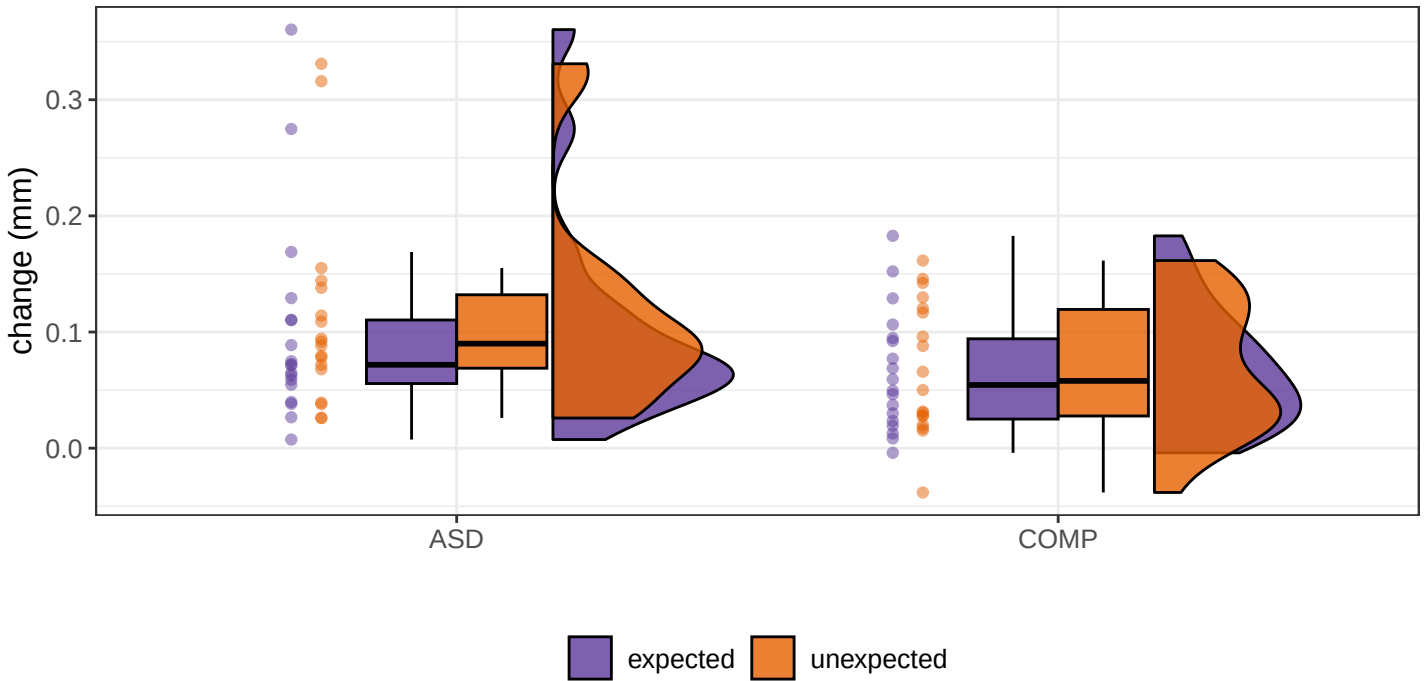
## S5.2 Pupil sizes

Modeled with a Gaussian linear mixed model, including the predictors group (ASD, COMP) and expected (yes, no) with their interaction as well as the covariate rts and the random intercept for subjects.

Table S11: Fixed effects of the pupil size model

|                      | Estimate | Est.Error | Q2.5   | Q97.5 |
|----------------------|----------|-----------|--------|-------|
| Intercept            | 0.085    | 0.013     | 0.059  | 0.109 |
| diagnosis1           | 0.014    | 0.013     | -0.012 | 0.040 |
| expected1            | -0.001   | 0.002     | -0.005 | 0.004 |
| rts                  | 0.000    | 0.000     | 0.000  | 0.000 |
| diagnosis1:expected1 | -0.002   | 0.002     | -0.006 | 0.003 |

Figure S9: Pupil size changes



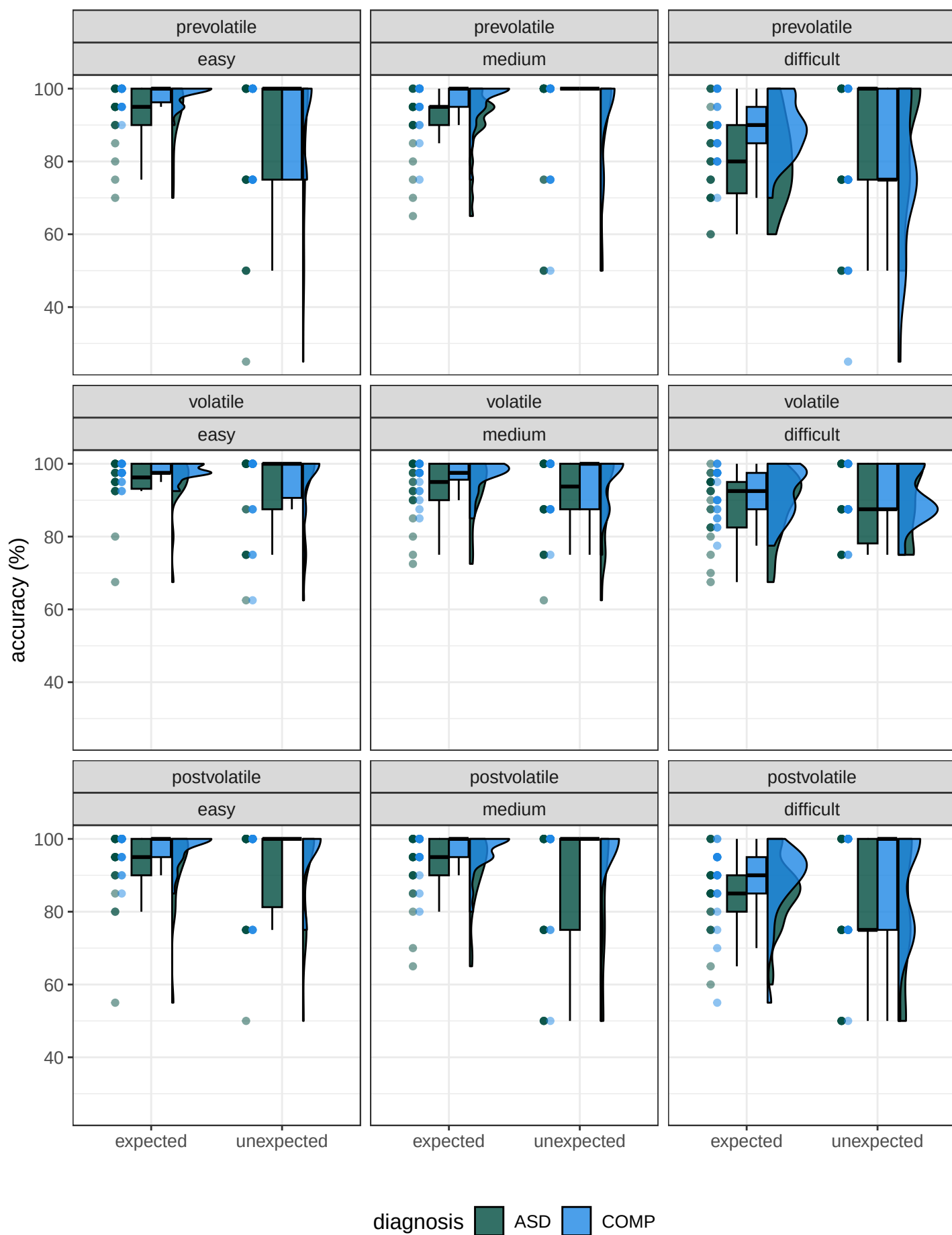
## S5.3 Accuracies

ANOVA of the ranked accuracies, including the predictors group (ASD, COMP), expected (yes, no), phase (prevolatile, volatile, postvolatile), difficulty (easy, medium, hard) and their interactions. Since this results in a large number of models, we only print the best 5 models. The full ANOVA is shared on GitHub: [https://github.com/IreneSophia/EMBA\\_PAL/blob/main/EMBA\\_PAL-ASD/\\_brms\\_models/aov\\_acc.rds](https://github.com/IreneSophia/EMBA_PAL/blob/main/EMBA_PAL-ASD/_brms_models/aov_acc.rds).

Table S12: ANOVA of ranked accuracies

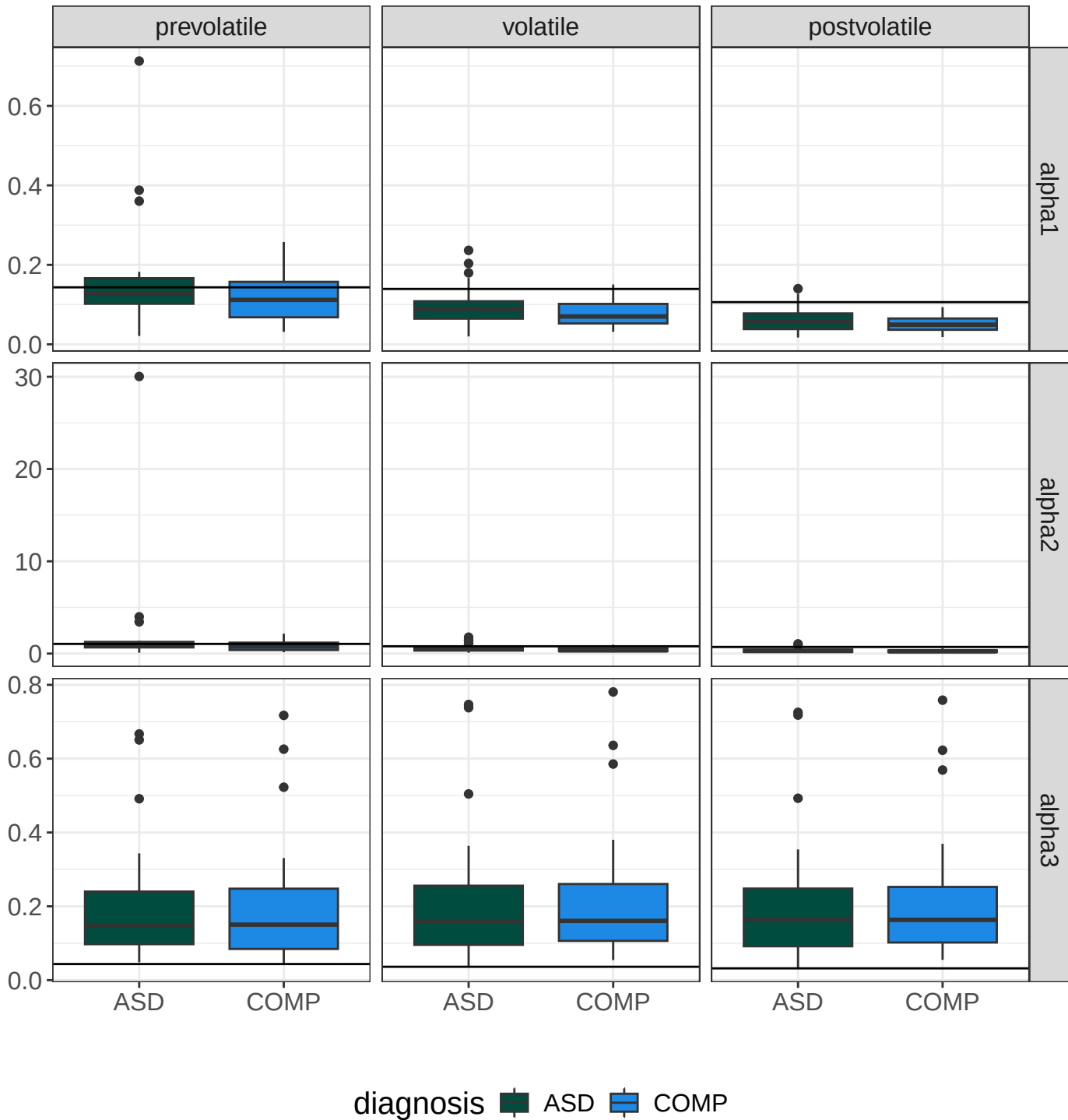
|                                                                               | bf    |
|-------------------------------------------------------------------------------|-------|
| diagnosis + expected + difficulty                                             | 41.15 |
| diagnosis + difficulty                                                        | 40.83 |
| diagnosis + expected + diagnosis:expected + difficulty                        | 40.29 |
| diagnosis + expected + difficulty + diagnosis:difficulty                      | 39.74 |
| diagnosis + difficulty + diagnosis:difficulty                                 | 39.39 |
| diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty | 38.93 |

Figure S10: Accuracies per subject



## S6 Learning rates proper

Figure S11: Learning rates



## S7 Handedness and button assignment

We counterbalanced whether the left or the right button was to be clicked to indicate a positive valence. Here, we check whether there were any differences between groups in terms of handedness and button assignment. Information on handedness of one participant is missing, because they skipped this question.

```
##                bf error                time                code
## Non-indep. (a=1) -2.444793      0 Sat Apr 25 14:07:10 2026 62fa3c5cd315
```

Table S13: Handedness distribution across groups

| diagnosis | handedness | n  |
|-----------|------------|----|
| ASD       | beidh ndig | 1  |
| ASD       | links      | 3  |
| ASD       | rechts     | 17 |
| ASD       | NA         | 1  |
| COMP      | links      | 3  |
| COMP      | rechts     | 19 |

Table S14: Handedness distribution across groups and key assignment

| diagnosis | key_pos | handedness | n  |
|-----------|---------|------------|----|
| ASD       | left    | beidh ndig | 1  |
| ASD       | left    | links      | 1  |
| ASD       | left    | rechts     | 9  |
| ASD       | right   | links      | 2  |
| ASD       | right   | rechts     | 8  |
| ASD       | right   | NA         | 1  |
| COMP      | left    | links      | 2  |
| COMP      | left    | rechts     | 9  |
| COMP      | right   | links      | 1  |
| COMP      | right   | rechts     | 10 |

## S8 Learning rates based on Bayesian Observer

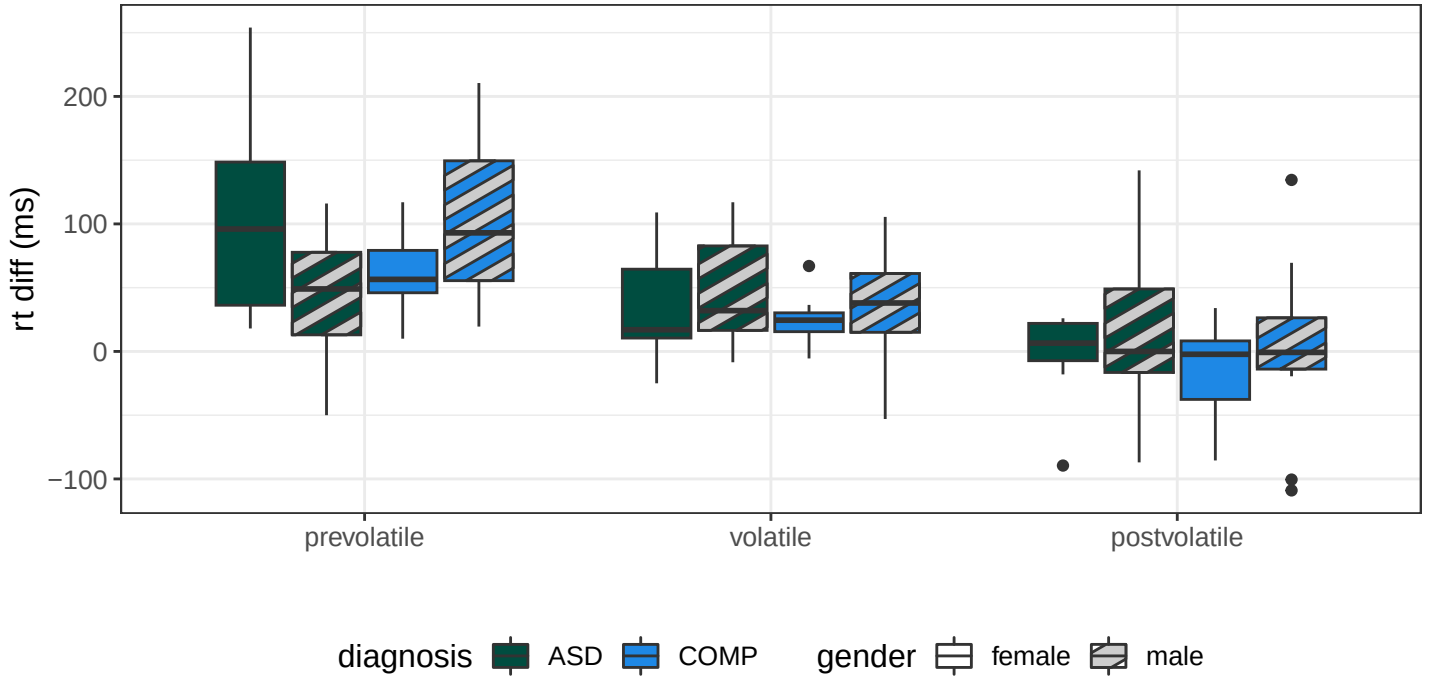
Table S15: Learning rate updates based on Bayesian Observer

| level                           | change                   | value |
|---------------------------------|--------------------------|-------|
| 2nd level learning rate updates | prevolatile to volatile  | 0.217 |
| 2nd level learning rate updates | volatile to postvolatile | 0.059 |
| 3rd level learning rate updates | prevolatile to volatile  | 0.004 |
| 3rd level learning rate updates | volatile to postvolatile | 0.001 |

## S9 Exploration of gender differences

### S9.1 Reaction times: expectancy effect

Figure S12: Expectancy effect in RTs including Gender



### S9.2 HGF parameters

Table S16: Fixed effects of the Phasic environmental volatility model including Gender

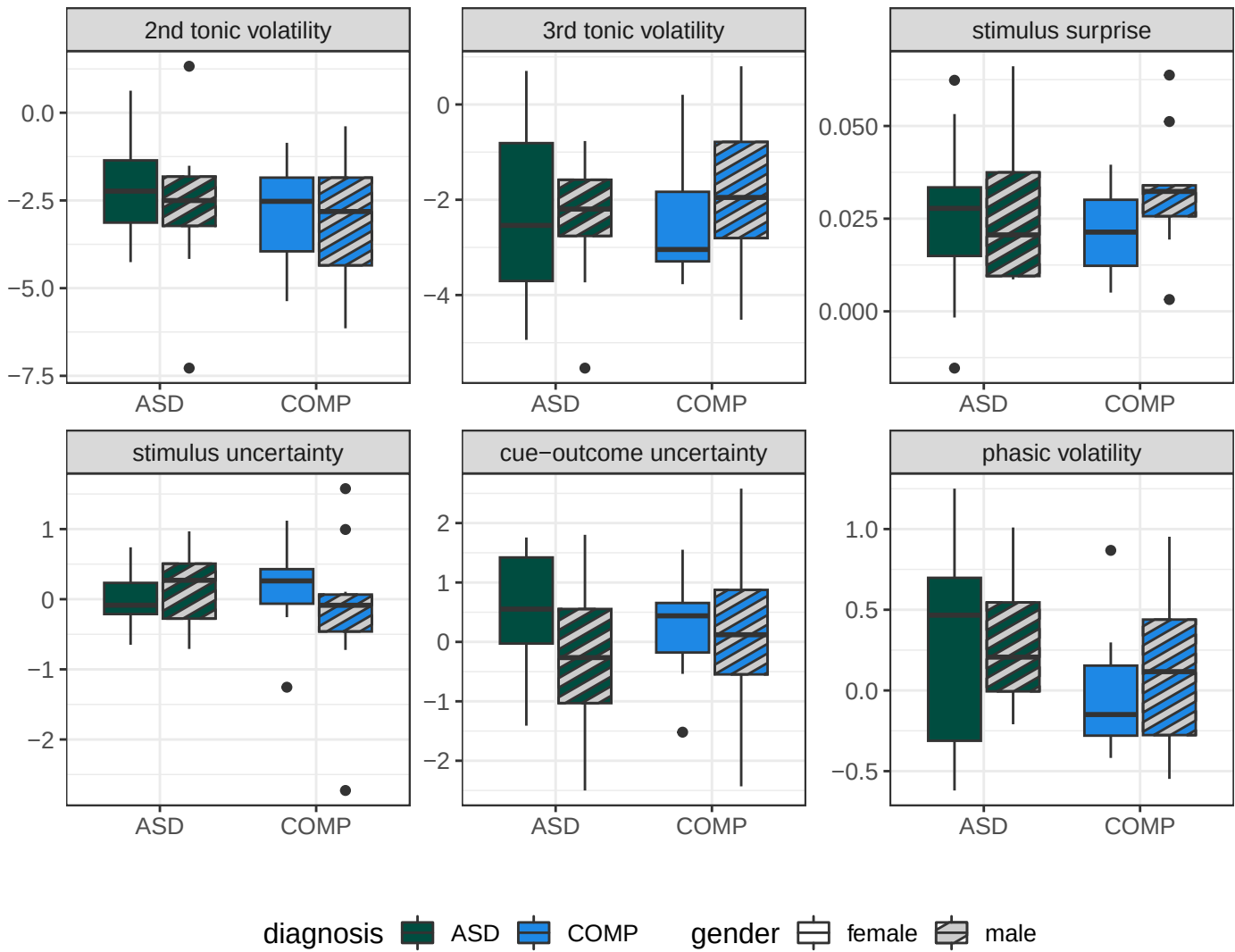
|                    | Estimate | Est.Error | Q2.5   | Q97.5 |
|--------------------|----------|-----------|--------|-------|
| Intercept          | 0.152    | 0.073     | 0.009  | 0.296 |
| diagnosis1         | 0.104    | 0.070     | -0.035 | 0.242 |
| gender1            | -0.043   | 0.070     | -0.180 | 0.096 |
| diagnosis1:gender1 | 0.014    | 0.071     | -0.123 | 0.153 |

Table S17: Fixed effects of the Tonic environmental volatility model including Gender

|                    | Estimate | Est.Error | Q2.5   | Q97.5  |
|--------------------|----------|-----------|--------|--------|
| Intercept          | -2.256   | 0.221     | -2.697 | -1.823 |
| diagnosis1         | -0.029   | 0.165     | -0.354 | 0.295  |
| gender1            | -0.068   | 0.164     | -0.390 | 0.256  |
| diagnosis1:gender1 | 0.100    | 0.166     | -0.228 | 0.423  |



Figure S13: HGF parameter



### S9.3 Learning rate updates

Table S18: ANOVA investigating learning rate updates including the predictor Gender (5 best models)

|                                           | bf      | bf.diff | bf.int                          |
|-------------------------------------------|---------|---------|---------------------------------|
| level + change                            | 105.331 | 0.010   | anecdotal evidence in favour of |
| level + change + level:change             | 105.321 | 0.630   | anecdotal evidence in favour of |
| diagnosis + level + change + level:change | 104.691 | 0.011   | anecdotal evidence in favour of |
| diagnosis + level + change                | 104.679 | 0.862   | anecdotal evidence in favour of |
| gender + level + change + level:change    | 103.818 | 0.040   | anecdotal evidence in favour of |
| gender + level + change                   | 103.777 | 0.124   | anecdotal evidence in favour of |

Figure S14: Learning rate updates

